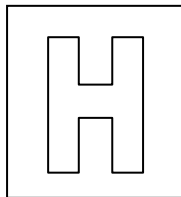


Candidate Name: _____

Class	Adm No



2021 Prelim Examination Pre-University 3

H2 Biology

9744/02

Paper 2 Structured Questions

13 September 2021

2 hours

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the question booklet.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question. At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
1	/13
2	/10
3	/11
4	/14
5	/8
6	/11
7	/11
8	/18
9	/4
Total	/100

This question paper consists of 28 printed pages, including 3 blank pages.

[Turn over

Answer **all** questions in this section.

1. The dopamine receptor D4 (Fig. 1.1) is a G protein-coupled receptor encoded by the *DRD4* gene on chromosome 11. D4 receptor is activated by the neurotransmitter dopamine.



Fig. 1.1

- (a)** Describe how the primary structure of D4 gives rise to its secondary structure.

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.....[2]

(b) Relate the structure of D4 to its role as a receptor protein.

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(c) The *DRD4* gene has a large number of alleles. Each allele has its characteristic length.

(i) Describe how gel electrophoresis reveals the differences between the different alleles of *DRD4* gene.

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- (ii) A scientist isolated the mRNA molecules from the cytoplasm of mouse cell and conducted gel electrophoresis to separate the mRNA molecules. Based on the number of bands which appeared on the gel, he reported that there are about 1,000 genes expressed in the cell.

Comment on the accuracy of this estimation.

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[Total:13]

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2. Bacteria contains an enzyme that catalyse the formation of PABA.

Fig. 2.1 shows the investigation on the effect of various concentrations of inhibitor on the production of PABA.

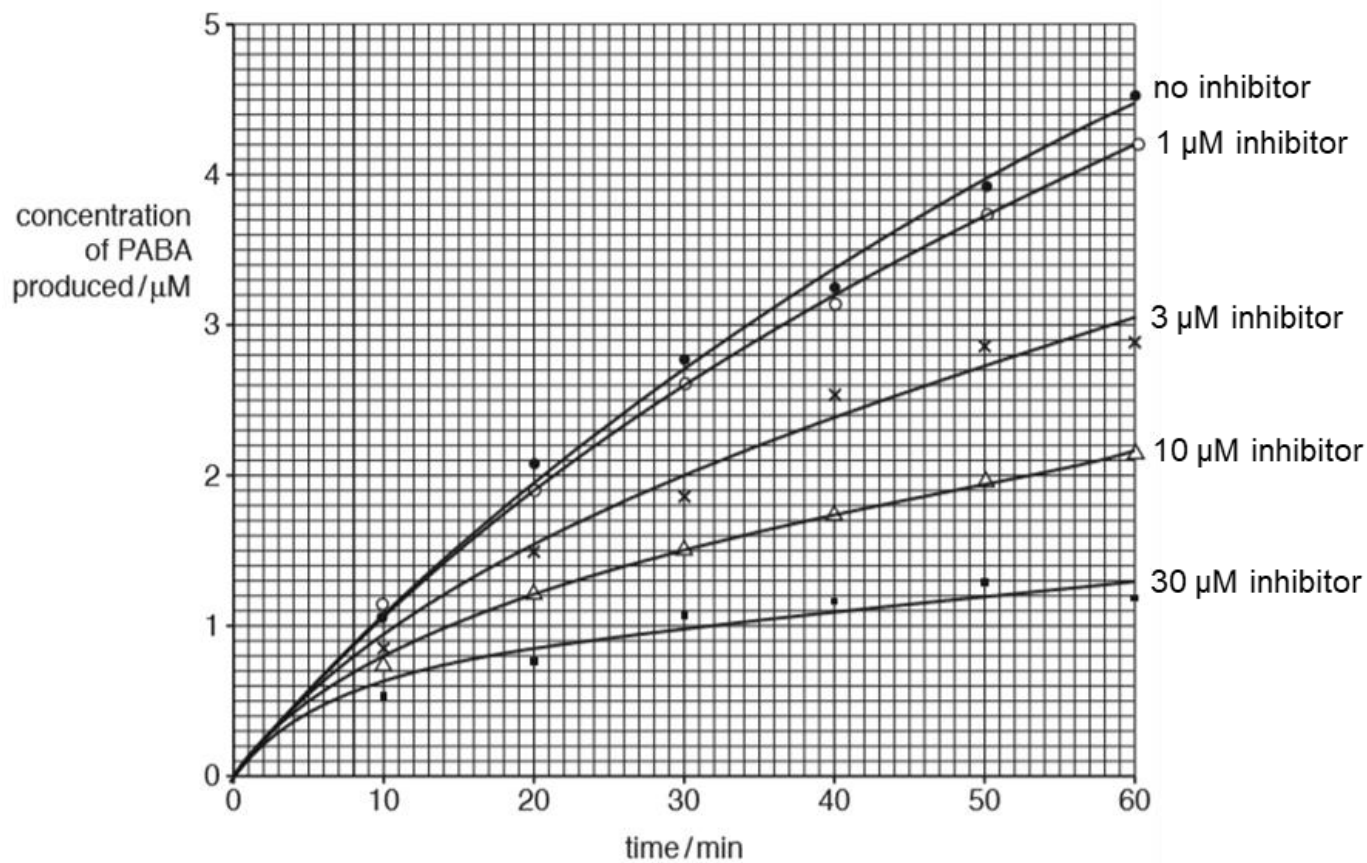


Fig. 2.1

(a) With reference to Fig. 2.1, describe the effect of inhibitor on the production of PABA.

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(b)

(i) Explain how a non-competitive inhibitor will affect the production of PABA.

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(ii) Explain why the results does not prove that the inhibitor works non-competitively.

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- (c) PABA is used to make folic acid. Folic acid is a molecule essential for the growth of most cells.

Bacteria cannot absorb folic acid from their surroundings. Folic acid from the diet is able to enter human cells but not able to cross bacterial cell walls. Human cells do not have an enzyme to make PABA.

Suggest why the inhibitor of this enzyme could be used as a drug to treat bacterial infections in humans.

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[Total: 10]

3. The two strands of a DNA molecule are held together by hydrogen bonds between complementary base pairs.

(a) Explain why the hydrogen bonding between the two strands of DNA is important for it to carry out its function.

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(b) DNA and RNA both contain nucleotide with adenine.

Complete Table 3.1 to compare

- DNA nucleotide with adenine
- RNA nucleotide with adenine
- ATP

Table 3.1

Feature	DNA nucleotide with adenine	RNA nucleotide with adenine	ATP
Contains nitrogen			
Contains a pyrimidine base			
Number of phosphate groups			
Name of the sugar component			

[4]

(c) Describe four differences between mRNA synthesis and polypeptide synthesis.

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[Total: 11]

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4. The graph in Fig. 4.1 shows how the number of chromosomes changes during the meiotic cell cycle of an organism. The three stages of interphase are shown as G_1 , S and G_2 . Meiosis I and Meiosis II are shown as M_I and M_{II} respectively. M_I and M_{II} each comprise of stages A, B, C and D.

At the start of interphase, the number of chromosome per cell is 4.

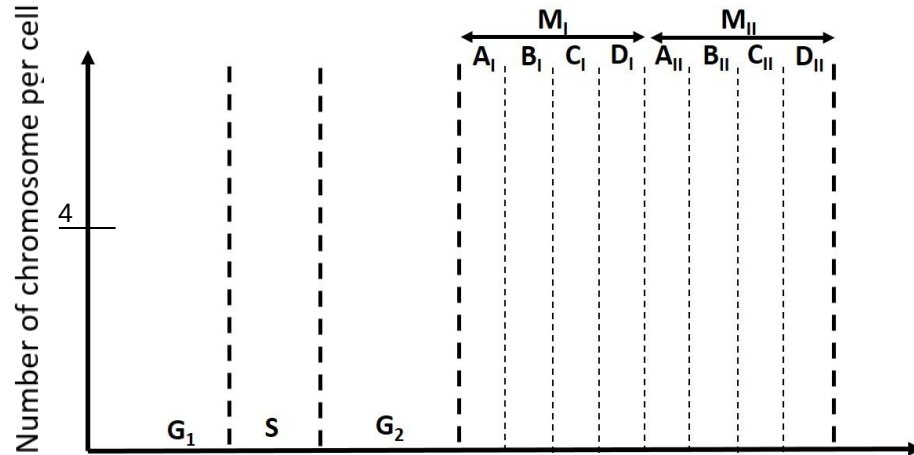


Fig. 4.1

- (a) On Fig. 4.1, draw, using line, how the number of chromosomes changes during meiosis. [2]

(b) Complete Table 4.1.

Table 4.1

Phase of Cell Cycle	Per cell	
	Number of DNA molecules	Amount of DNA(arbitrary units)
G₁ phase	4	2x
G₂ phase		
A_I		
D_I		
A_{II}		
D_{II}		
Cytokinesis	2	x

[2]

(c) Account for the changes in the number of chromosome and the amount of DNA per cell from G₁ till end of D_I.

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(d)

- (i)** Dysregulation of cell cycle can lead to cancer. In individuals with cancer, such as breast cancer, not all the cells in the patient's breast are transformed into cancer cells.

Explain why.

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- (ii)** Tamoxifen is a type of drugs used to treat breast cancer. It acts by blocking any estrogen present in the body from attaching to the estrogen receptor on the cancer cell. This eventually slow or prevent growth of tumours.

Suggest the normal function of estrogen in the human body.

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..... [1]

Upon binding of estrogen to estrogen receptor, estrogen receptors move from the cytoplasm into the nucleus.

- (iii)** Explain why estrogen receptor moves into the nucleus.

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[Total: 14]

5. Coronaviruses are a group of viruses with crown-like spikes on their surface. There are four main sub-groupings of coronavirus, and one of them is the β -coronavirus. In recent studies, scientists found evidence that coronaviruses are able to hijack lysosomes and use it as an exit route from the cells.

To show that β -coronavirus do not rely on normal pathway in exiting the host cell, the scientists used Brefeldin A (BFA), an inhibitor which prevents protein transport between rough endoplasmic reticulum and golgi complex.

The results of their investigation is shown in Fig. 5.1.

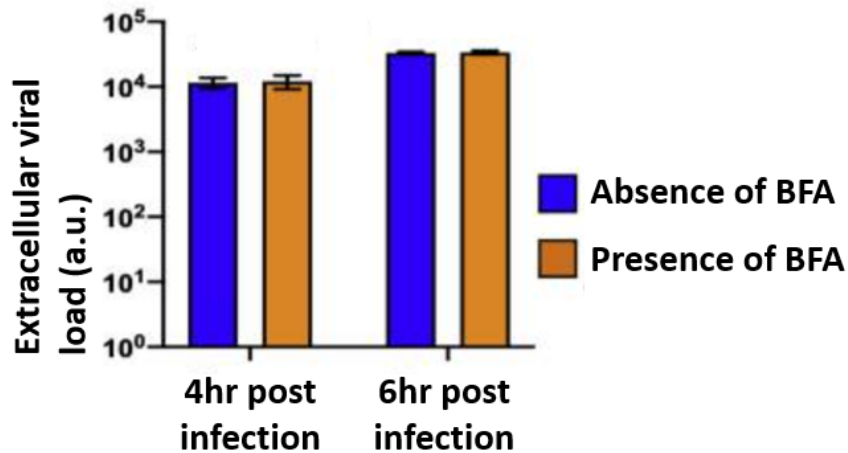


Fig. 5.1

- (a) With reference to Fig. 5.1 and the information provided, justify how the use of BFA provided evidence that β -coronavirus do not rely on the normal pathway to exit the host cell.

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- (b)** LAMP1 is a protein that is usually found on the membrane of lysosomes. It was observed that cells infected with β -coronavirus had elevated levels of LAMP1 on the cell surface membrane.

Explain how this further supports the hypothesis that coronavirus exits the cell via lysosomal trafficking.

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- (c)** Suggest why the scientists measured the amount of virus in the extracellular environment.

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- (d)** The team of researchers hypothesised that coronavirus were able to 'deacidify' the internal environment of lysosomes to disrupt lysosomal functions.

- (i)** Suggest how coronavirus might be able to 'deacidify' lysosomes.

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- (ii) Suggest why it is not recommended to inhibit lysosome as part of anti-viral treatment.

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[Total: 8]

6. *Staphylococcus aureus* (*S.aureus*) is a bacteria that is found on human's skin. Methicillin resistant *Staphylococcus aureus* (MRSA) is a variety that is difficult to treat because of its resistance to methicillin and some other antibiotics.

Fig. 6.1 shows the total number of *S.aureus* infection in children under 15 years old, and the percentage of MRSA (relative to all *S.aureus* infection) in England and Wales from 1990 to 2001.

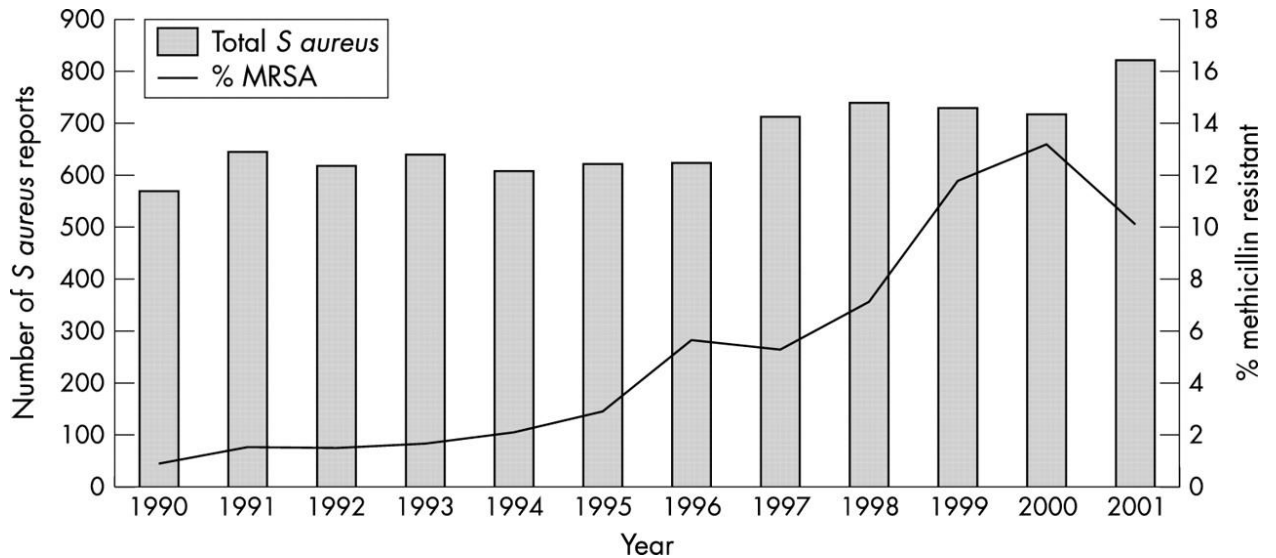


Fig. 6.1

- (a) A student claimed that the increase in percentage of MRSA is due to increase in total number of *S.aureus*.

With reference to Fig. 6.1, justify this claim.

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(b) Methicillin is a semi-synthetic penicillin.

Describe how methicillin kills bacteria.

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(c) A strain of the bacterium *Pseudomonas aeruginosa*, strain R, has a gene coding for the pump and is resistant to a group of antibiotic known as β -lactam.

The minimum inhibitory concentration (MIC) is the lowest concentration of antibiotics that prevents bacteria colony from forming. Two mutant strains, strain 1 and strain 2 are derived from strain R. The MICs of the β -lactam for strain R, mutant 1 and mutant 2 were determined. Mutant 1 and 2 differ from strain R in the expression of the gene coding for the pump.

The MICs for the three strains of *Pseudomonas aeruginosa* are shown in Table 6.1.

Table 6.1

Strain of <i>Pseudomonas aeruginosa</i>	MIC of β -lactam ($\mu\text{g cm}^{-3}$)
Resistant strain R	65
Mutant strain 1	0.5
Mutant strain 2	270

With reference to Table 6.1,

(i) explain why the MICs for mutant strains 1 and 2 differ from that for strain R.

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(ii) explain how a population of strain R could be replaced by mutant strain 2.

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[Total: 11]

7. The Hawaiian archipelago is a group of volcanic islands in the Pacific Ocean. Fig. 7.1 shows the relative locations of five of these islands.



Fig. 7.1

Table 7.1 shows the size and age of these five islands and the total number of *Mecyclothorax* ground beetle species and their species density, on each island.

Data for the island of Maui is shown as two distinct regions, West and Haleakalā. This is because they formed at different times from two separate volcanoes.

Table 7.1

island	area / km ²	age of island / million years	total number of <i>Mecyclothorax</i> species	species density / number of species per km ²
Hawai'i	10 433	0.4	30	0.003
Lāna'i	364	1.3	3	0.008
Maui (West)	443	1.3	27	0.061
Maui (Haleakalā)	1440	1.1	116	0.081
Moloka'i	673	1.9 – 1.8	43	0.064
O'ahu	1545	3.7 – 2.6	20	

(a)

(i) Complete Table 7.1 by calculating the density of *Mecyclothorax* beetle species on the island of O'ahu. [1]

(ii) Using Table 7.1, suggest why the island of Hawai'i has the lowest density of *Mecyclothorax* beetle species.

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 [2]

(iii) Using Table 7.1 and Fig. 7.1, suggest why O'ahu has a lower number of *Mecyclothorax* beetle species than Moloka'i.

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 [1]

Biological species concept is one of the species concept used to define a species.

(b) State two limitations of biological species concept.

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 [2]

- (c) The rate of speciation of *Mecyclothorax* beetles on the slopes of the volcano Haleakalā, on Maui, is the highest recorded for any genus or location on earth.

The volcano last erupted 400 years ago. This produced lava flows which cut through ancient forest and vegetation, dividing it into many separate microhabitats.

Explain how the large number of *Mecyclothorax* beetle species on Haleakalā developed.

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[Total:11]

8. Fig. 8.1 shows the series of events that occur when naïve B lymphocytes encounter its specific antigen.

Upon binding to specific antigen, B lymphocyte is activated and undergoes cell growth. Activated B lymphocytes undergo mitosis to form a pool of B lymphocytes.

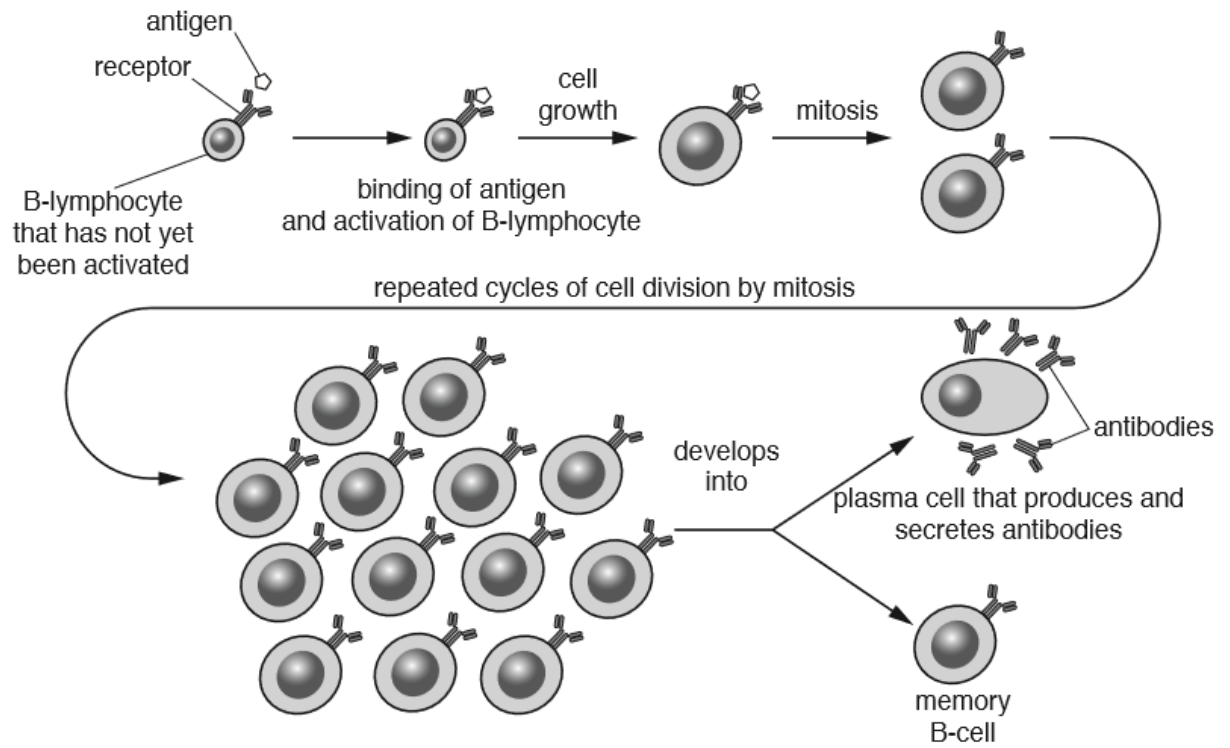


Fig. 8.1

- (a)** Explain the importance of forming a pool of B lymphocytes.

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The pool of B lymphocytes differentiate into either plasma cells or memory B cells.

- (b)** Explain how B lymphocytes differentiate into memory B cells, and describe the roles of memory B cells.

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- (c)** Antibodies are glycoproteins.

- (i)** Name and describe the structure of the organelle involved in the synthesis and chemical modification of antibodies.

Name

Structure

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(ii) Suggest why the first antibody synthesised is not secreted.

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(iii) The genes responsible for antibody production are found on different chromosomes, such as chromosomes 2 and 14 in humans.

Explain how one antibody molecule is the product of more than one gene.

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(iv) Human has the ability to make millions of different antibodies. Class switching is one of the processes that give rise to the different antibody molecules.

Explain how class switching gives rise to the many variety of antibodies.

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[Total:18]

9. Dengue fever is a mosquito-borne viral disease and a regular epidemic in Thailand. The peak of the dengue epidemic period is around June to August during the rainy season. It is believed that climate is an important factor for dengue transmission. Fig. 9.1 shows how changes in temperature affects the average monthly incidence rate of dengue cases in Chiang Rai, Thailand from 2004 – 2014.

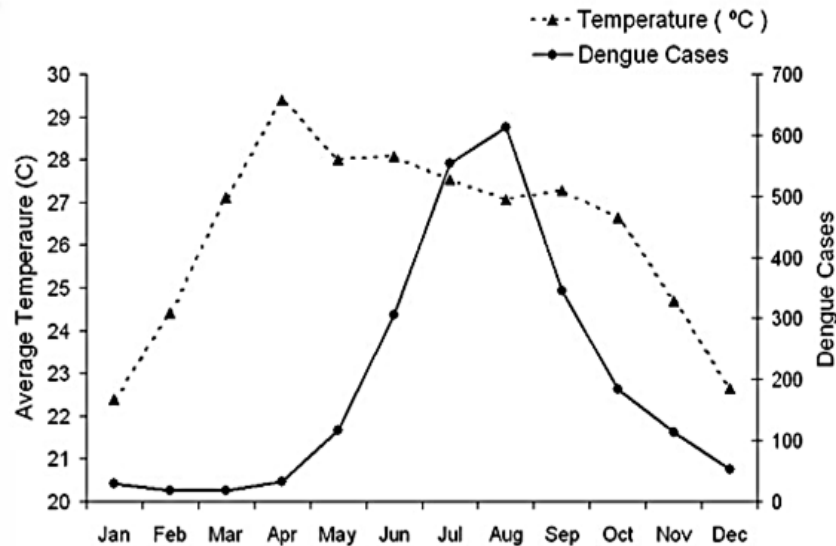


Fig. 9.1

- (a) State how each data point in Fig. 9.1 is obtained.

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- (b) With reference to Fig. 9.1, explain the trend of incidence of dengue cases with regard to changes in temperature.

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[Total:4]

End of Paper